Open-Source Modeling of Photoluminescence Dynamics in Semiconductor Quantum Dots

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1 Mentors

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2 Project Description & Objectives

Semiconductor quantum dots (QDs) are nanoscale systems with discrete electronic states governed by quantum confinement effects. Due to their tunable optical properties and relevance in quantum photonics, they are promising candidates for deterministic single-photon sources and semiconductor-based qubits.

This project aims to develop an open-source computational toolkit in Python for simulating photoluminescence (PL) spectra and exciton dynamics in semiconductor quantum dots.

The core objectives are:

1. Implement analytical confinement models such as the Brus and Kayanuma equations for calculating size-dependent energy levels.
2. Simulate exciton population dynamics, radiative decay, and decoherence using Lindblad master equation approaches with QuTiP.
3. Generate synthetic photoluminescence spectra including homogeneous and inhomogeneous line broadening mechanisms.
4. Develop an interactive terminal-friendly or lightweight dashboard for exploring structural and environmental parameter effects.
5. Provide educational Jupyter Notebook tutorials demonstrating quantum confinement and open quantum system dynamics.

3 Prerequisites for Interns

- Proficiency in Python and familiarity with NumPy/SciPy.
- Basic understanding of quantum mechanics (Hamiltonians, bra-ket notation).
- Interest in computational physics or open quantum systems.
- Prior experience with QuTiP is helpful but not required.

4 Work Plan (1 July – 15 August 2026)

Weeks 1–2 (July 1 – July 12): Literature Review & Setup

- Team onboarding and Git workflow setup.
- Review of semiconductor quantum dot theory and photoluminescence mechanisms.
- Familiarization with QuTiP and numerical simulation workflows.

Weeks 3–4 (July 13 – July 26): Core Simulator Development

- Development of confinement energy calculation modules.
- Construction of excitonic Hamiltonians.
- Implementation of Lindblad master equation solvers for decoherence and relaxation dynamics.

Weeks 5–6 (July 27 – August 9): Spectral Synthesis & Validation

- Generation of theoretical photoluminescence spectra.
- Implementation of line broadening models.
- Validation against theoretical predictions and benchmarking.

Week 7 (August 10 – August 15): Documentation & Final Presentation

- Preparation of documentation and tutorials.
- Development of Jupyter notebook demonstrations.
- Compilation of final report and presentation.

5 Deliverables

1. Open-source Python library for simulating quantum dot photoluminescence spectra and exciton dynamics.
2. Jupyter Notebook tutorials covering:
   - Quantum confinement
   - Exciton decay
   - Open quantum system simulations
   - Spectral generation
3. Visualization and parameter exploration tools.
4. Final technical report and presentation for QIntern 2026.

6 Technologies

- Python
- NumPy
- SciPy
- Matplotlib
- QuTiP
- Jupyter Notebook
- Git/GitHub

7 Expected Impact

This project bridges semiconductor physics and open quantum system modeling while promoting open-source scientific computing in quantum technologies education and research. It provides a foundation for further exploration of quantum photonics systems and quantum dot-based devices.